

Jason Cabrejos

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Experience

Burlington Northern Santa Fe – BNSF Railway

March 2023 – Present

Robotic Embedded SW Engineer – Full Time Contractor

- Implemented a low-cost real-time inventory tracking system for under \$10K per unit merging technologies such as IoT, Networking, Power over Ethernet (PoE) systems, and edge computing solutions using the **Nvidia Jetson**
- Achieved 92% accuracy on a **production scale** by integrating GPS, RTK, and Machine Vision models saving 5 million dollars per year
- Scaled to multiple hardware stacks around the nation using **Docker** and **Azure IoT Hub** to manage our fleet of devices

Automation and Intelligent Systems Lab – UTA Research Institute

October 2022-February 2023

Researcher

- Developed Baxter robotic arm scripts using **ROS** and **OpenCV** for object recognition and retrieval in community engagement projects
- Enhanced performance in industrial settings by overhauling sensor stability and **reducing vibration induced errors** on multiple platforms
- Proficient in Rapid Prototyping techniques (3d Printing, lathe, CNC and Manual Milling operations) to reduce development time

Crown Castle

January 2022 – August 2022

Robotic Development Intern

- Reduced personnel required for a Fiber-Optic deployment operation by 30% through automation of industrial equipment using **Robotic Operating System (ROS)**, **Python** and **Ubuntu Linux**
- Established communication between an industrial vehicle's CAN bus and an embedded **Nvidia Jetson**, enabling sensor feedback and actuator control leading to automated curb following and cable drift correction
- Conducted research on industrial IP-rated components ensuring its compatibility with preexisting systems for wide deployment

Laboratory of Advanced Materials Manufacturing and Analysis

June 2021 – January 2022

Researcher

- Designed an Automated Fiber Placement (AFP) tool to be mounted on a Kuka robotic arm for experimental composite layups
- Developed software to enable the Automated Fiber Placement tool to communicate with the Kuka ecosystem and **PLC's** using **EtherNet/IP** and **Python** resulting in the creation of a slicing software that was able to directly control the AFP Tool and robot arm
- Oversaw the construction of a work cell ensuring it complied with all safety requirements and met the project's specifications

Leaderships, Projects & Awards

Hackathons

Various Times

Finalist Placements

- Improved child safety by pitching a camera system monitor activity without the driver needing to take their eyes of the road – 1st place
- Presented a Machine Learning model and business plan to predict potential failures on the railway using big data – 1st place
- Trained a mock interviewer LLM using **Microsoft Copilot Studio** and **Power Automate** to help practice and get feedback -2nd place
- Trained a **Tensorflow Computer Vision** model to automate the recognition and sorting of waste using a robot– 3rd place
- Agricultural IoT platform to determine when to water/harvest using **Computer Vision**, **Azure**, and a **Raspberry Pi** network – 3rd place

Rover Team

August 2017 – August 2021

President, Arm Leader, Member

- Led a team of 30 to the University Rover Challenge, traversed terrain, retrieved objects, and searched for life - placed 15th out of 88 teams
- Managed a team of 5 through the design, manufacturing, and assembly of a 5 DOF Robotic Arm over the course of an academic year
- Utilized **Solidworks** and **Ansys** for modeling and Finite Element Analysis (FEA) Simulations to reach a factor of safety of 2.4
- Developed a PID Joint Space Controller using **AutoLev**, simulated on **Matlab** and ran on **Arduino** with an accuracy of ± 0.4 in

NASA L'Space Academy

May 2019 – November 2019

Project Manager

- Managed 2 projects: a submarine to traverse Saturn's moon Titan and improving robotic systems on the International Space Station
- Coordinated with 6 students from across the nation to achieve deadlines, share knowledge, and foster a culture of innovation
- Gained insights into NASA's approach to teamwork, workflow, planning, and budgeting in dynamic engineering setting

Education

University of Texas at Arlington

August 2025

- Master of Science in Computer Science
 - Focus on Intelligent Systems and Computer Networks
- Bachelor of Science in Mechanical Engineering
 - Minor in Computer Science

Qualifications

- Programming Experience: Python, C++, Matlab, ROS, C#, Labview
- Tools and Technologies: Git, Docker, Rabbitmq, Tensorflow, Roboflow, MongoDB, Azure, AWS, IOT
- Modeling Experience: Solidworks (Certified Associate (CSWA) – Issued 2016), Solid edge